THE FORMAL COMPLETION OF A FORMAL ALGEBRAIC SPACE ALONG A CLOSED SUBSET IS A FORMAL ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the functor whose being a formal algebraic space is asserted; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definition of a formal algebraic space through a covering, and etaleness of the base changes of a representable etale morphism. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 8012-8014, and Spaces, Lemma spaces-lemma-base-change-representable-transformations-property, invoked at proof lines 8014-8017. Provide the covering whose pieces reduce the statement to the affine case.
- The completion of an affine formal algebraic space along a closed subset of its reduction is affine. Same chapter, Lemma lemma-completion-affine-formal-is-affine-formal, invoked at proof lines 8023-8025. Gives the affine case, which is the last step.
- Surjectivity of a map of sets underlying an fppf covering, and the sheaf property of the completion functor. Standard, used at proof lines 8006-8010. Establishes that the functor is an fppf sheaf.

Source reference. Original lemma 0GVQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7969–7969. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02WL: sources/stacks-chow/tags.txt, source lines 3808–3808. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be a formal algebraic space over S. Let $T \subset |X_{red}|$ be a closed subset. Then the functor*

$$X_{/T} : (Sch/S)_{fppf} \longrightarrow Sets, \quad U \longmapsto \{f : U \to X \mid f(|U|) \subset T\}$$

is a formal algebraic space.

Proof. The functor $X_{/T}$ is an fppf sheaf since if $\{U_i \to U\}$ is an fppf covering, then $\coprod |U_i| \to |U|$ is surjective.

Choose a covering $\{g_i : X_i \to X\}_{i\in I}$ as in Definition 0.1. The morphisms $X_i \times_X X_{/T} \to X_{/T}$ are étale (see Spaces, Lemma 02WL) and the map $\coprod X_i \times_X X_{/T} \to X_{/T}$ is a surjection of sheaves. Thus it suffices to prove that $X_{/T} \times_X X_i$ is an affine formal algebraic space. A U-valued point of $X_i \times_X X_{/T}$ is a morphism $U \to X_i$ whose image is contained in the closed subset $|g_{i,red}|^{-1}(T) \subset |X_{i,red}|$. Thus this follows from Lemma 0GVQ. □